

Mobile IP

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Need of MIP

The Internet Protocol (IP) routing is based on the IP addresses that define where the destination node should be found. Moving the host with a specific IP address generates problems, since the routing algorithms rely on using network prefixes. This means that all hosts having the same first bits in the address are supposed to be found in the same part of the network. If the IP address of a host is changed when the host has active connections, it leads to breaking of established connections.

Drawbacks of MIPv4

Despite providing the session continuity, unfortunately the MIP mechanism has some serious drawbacks.

□ **Triangular Routing:** As illustrated in Figure 1, when the MN is moved to the foreign network then in order to communicate, the datagrams are first routed from CN to HA then HA send the data packets to MN. However, for the data packets transfer from MN to CN the datagrams can be forwarded directly from MN to CN. The additional routing introduced by this triangular routing mechanism highly increases end to end delay compared to optimal routing which is not acceptable for the delay sensitive applications like VoIP, etc. It has been reported that in case of datagrams transmission from CN to MN within a campus network, the MIP increases end to end delay by 45%. Moreover, the end to end delay is expected to be increased when the two entities are physically far apart (for example, in case of WAN).

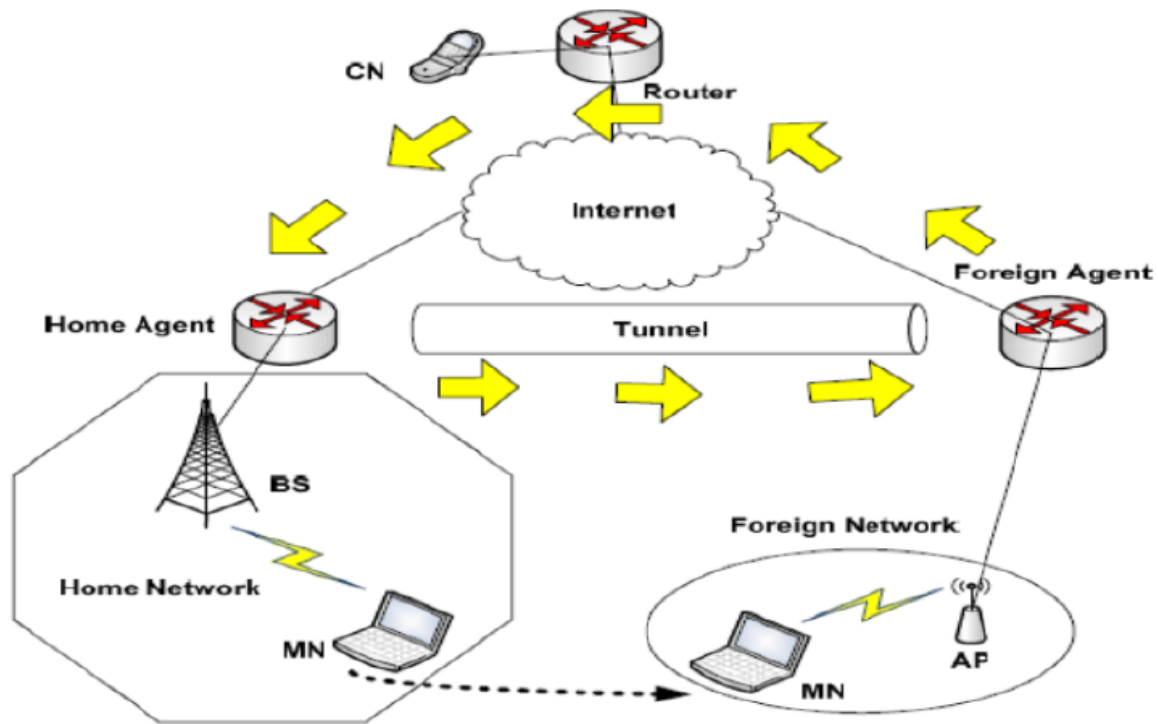


Figure 1: Triangular Routing

□ **Additional Protocol Overheads:** When the MN is moved to the foreign network then the HA tunnels the data packets to the MN's new point of attachment. Introduction of additional encapsulation overheads while establishing the data tunnel is one of the major drawbacks of MIP. The IP-in-IP encapsulation significantly adds the overheads which is particularly very critical for the real time services. The encapsulation is performed in a manner that the original datagram is wrapped with the outer IP envelope. Consequently, the final packet contains the outer IP header as well as the inner datagram. Typically, 20 bytes overhead is appended to every data packet which simply wastes the network available bandwidth. Furthermore, slow down the processing speed of intermediate routers and network devices. Therefore, in addition to the triangular routing, additional encapsulation overheads are another factor that further increases the network latency.

□ **Seamless Mobility Lacking:** Seamless mobility is the prime objective when designing any mobility management protocol for the integrated network design. It has been reported by several studies that the long duration of the MIP handoff process significantly increases the high handoff latency and packet loss during handoffs. Consequently, MIP is unable to provide seamless mobility.

□ **Additional Network Components:** As discussed earlier, the MIP mechanism requires to install the mobility agents i.e., HA and FA in the home and foreign networks, respectively. Introduction of mobility agents is considered as the shortcoming of MIP approach, since, it increases the cost of network deployment.

□ **Single point of failure:**

Since, all data are traversing through the HA, therefore, HA can be a single point of failure. In addition, in case of increasing mobile devices HA can become the network performance bottleneck.

These limitations bring the researchers' attention towards the MIPv6 route optimization mechanism.

MIPv6 Route Optimization Mechanism

Mobile IP version 6 is the mobile extension to IP version 6. The MIP fundamental principle is that a mobile node should use two IP addresses, a permanent address (the home address, assigned to the host and acting as its global identifier) and a temporary address (the care-of address CoA, providing the host's actual location). MIPv6 retains the general ideas of home network, home agent, and care-of address from MIPv4. However, it has a slightly different philosophy and a much-improved design than its predecessor.

While a mobile node is attached to its home network, it is able to receive packets destined to its home address, and being forwarded by means of conventional IP routing mechanisms. When the mobile node moves into a new network (visited/foreign network), its movement is detected and a new association is made with mobility agents (foreign agents) in the new domain. In MIPv6 a mobile node is more "independent" and does not have to rely on an access router to obtain a CoA and register with the home agent.

Despite many similarities with the MIPv4, the major advantage of the MIPv6 route optimization mechanism is the avoidance of several shortcomings of MIPv4. For example, FA deployment is not required. Moreover, the IP-in-IP encapsulation and triangular routing are avoided by introducing the route optimization technique. As illustrated in Figure 2, in contrast to the MIPv4 mechanism and bidirectional tunneling mechanisms in which the packets are routed from the HA, the MIPv6 route optimization allows the shortest path communication between the mobile devices by enabling the CN and the MN to send the data packets directly to each other. Consequently, the congestion in the home network and HA can be eliminated.

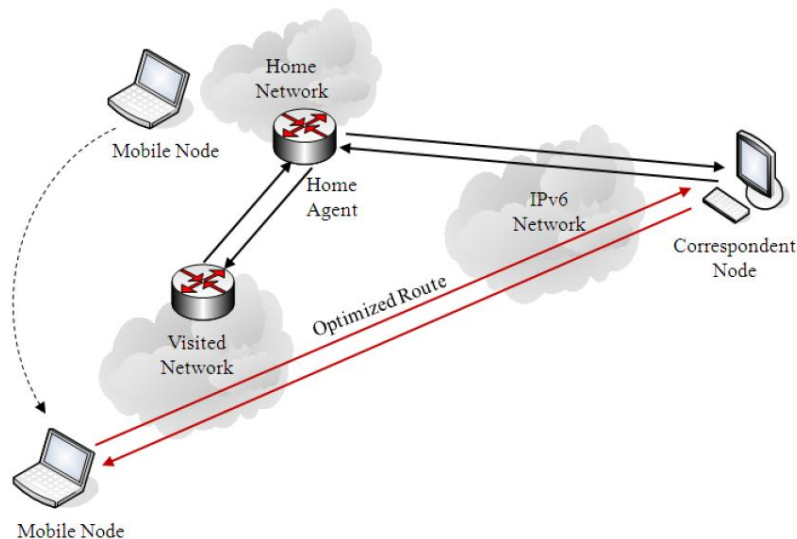


Figure 2: MIPv6 Route Optimization Mechanism

To establish a direct communication path, the MN not only sends the registration request to the HA, moreover, the CN is also updated with the MN relocation. Since, the CN is aware with the current CoA of the MN after relocation, therefore, instead of sending data packets to the HA, the CN establishes a direct communication path with the MN. This direct communication mechanism helps to reduce the end to end delay.

MIPv6 Handoff Operation/ Timing sequence

The handoff latency of a roaming MN can be measured as the time when MN stop sending/receiving packets from the previous access network to the time when the MN starts sending/receiving data packets to the new access network. The Mobile IPv6 vertical handoff operations are performed as illustrated in Figure 3.

When the MN moves and detects the foreign network, it performs Route Discovery/ Movement Detection (RD/MD) mechanism. The RD contains the handshake of Route Solicitation (RS) message sent by the MN, which is responded back with the Route Advertisement (RA) message of the foreign network access point. In order to configure the CoA address, the MN extracts the foreign network prefix information from RA. Based on the network prefix information and MAC address of MN, a CoA is generated by the MN.

According to the IPv6 specifications, the Duplicate Address Detection (DAD) procedure is required to perform to ensure the uniqueness of the configured CoA. The DAD scheme is performed by the MN by sending the Neighbor Solicitation (NS) to the new network and wait for the Neighbor Advertisement (NA) message. If no NA is received in response of NS for the time of 1000ms, then according to the IPv6 recommendation, the configured CoA is considered unique. After the successful unique CoA configuration, the CoA registration procedure is performed by the MN with the HA and CN. To perform the registration procedure to update the relocation information of MN, Binding Update (BU) messages are sent by the MN to HA and CN which are acknowledged by Binding Acknowledge (BA) messages. It is worth remaindering that the Return Routability (RR) is performed between the HA and CN update procedure. Afterwards, all the data packets are routed to the MN at its new location.

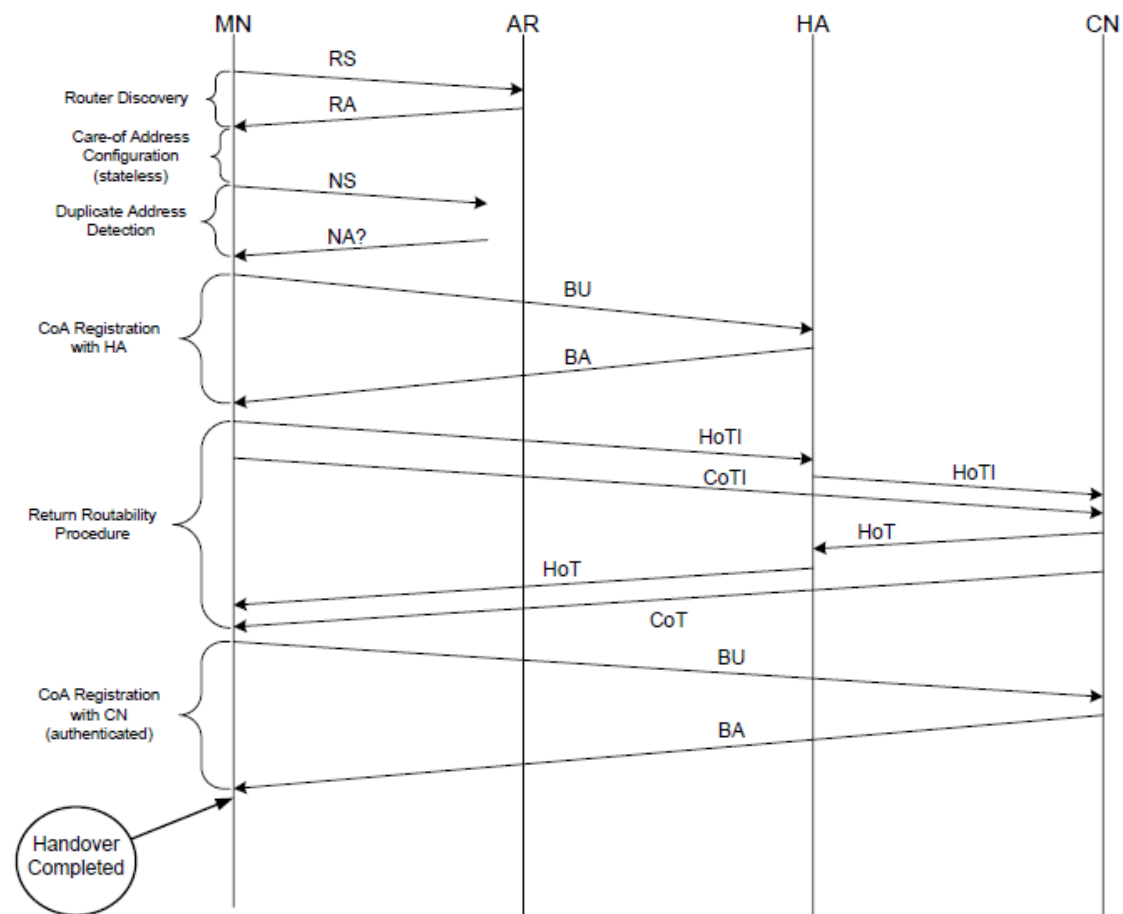
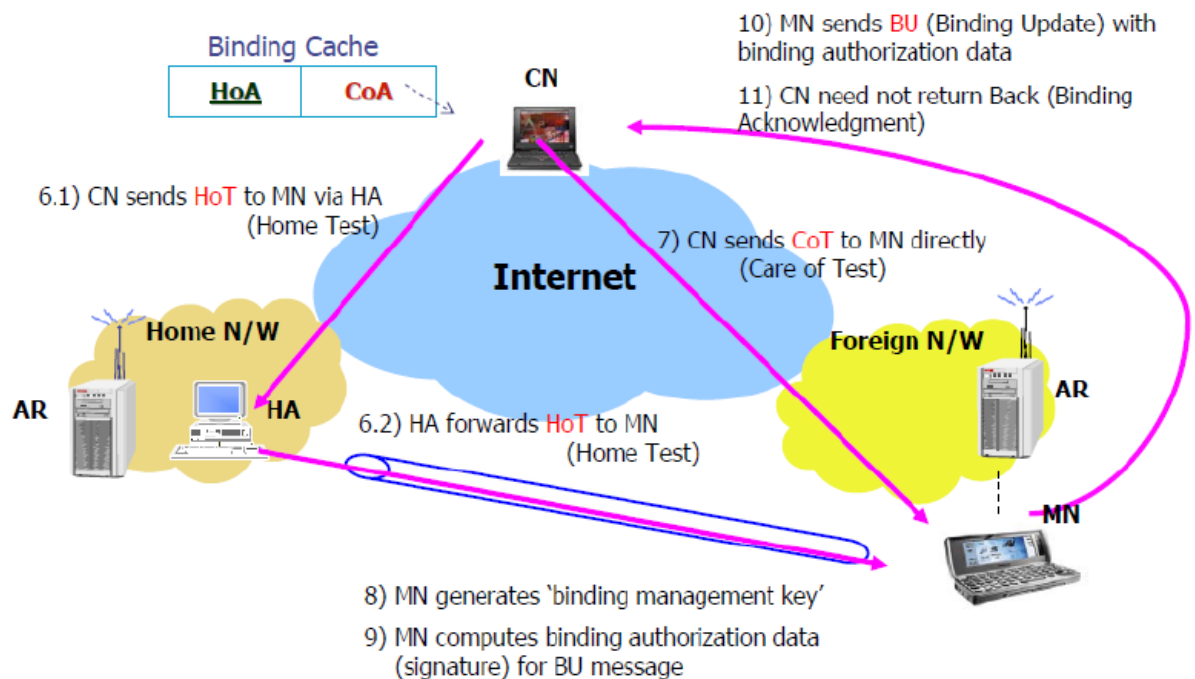
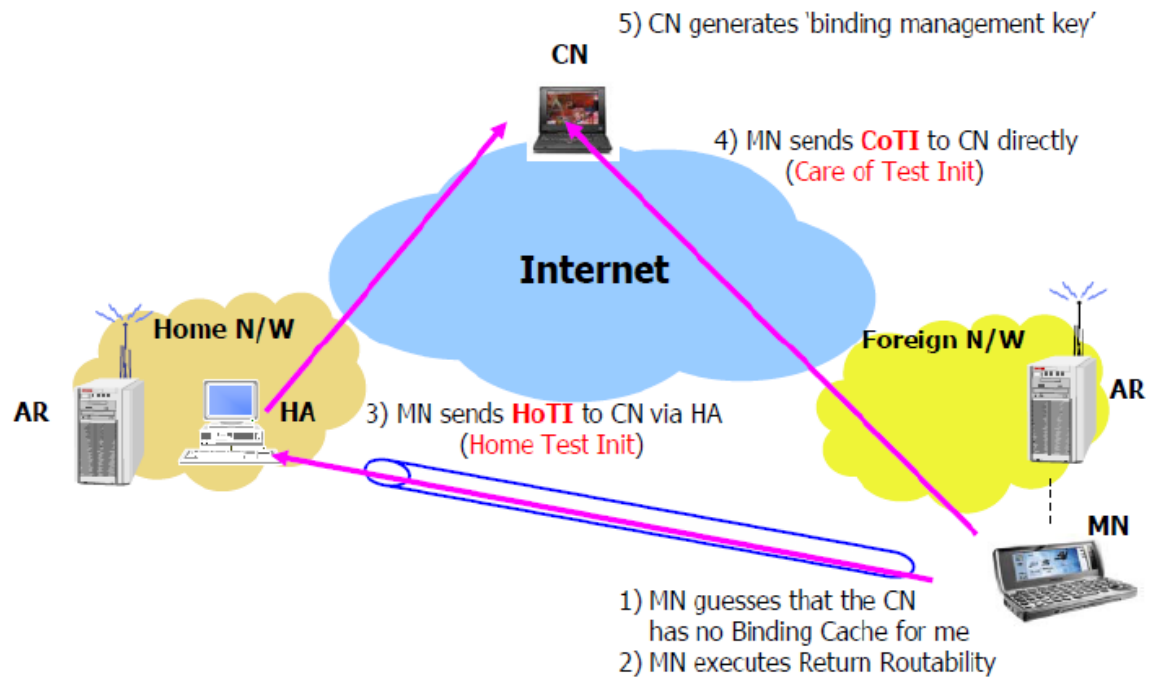


Figure 3: MIPv6 signaling during handoff

Return Routability Procedure

Actually, after establishing a binding update with the HA the MN initiates a return routability procedure with the CN. The procedure involves four messages. The Home and Care-of Test Init messages (HoTi and CoTi) are sent from the MN to the CN at the same time, the first one using the HA as intermediate router. The procedure requires very little processing at the correspondent node, and the Home and Care-of Test messages (HoT and CoT) can be returned quickly, perhaps nearly simultaneously. The purpose of this procedure is to enable the CN to make sure that the mobile node is in fact addressable at its claimed care-of address as well as at its home address. Only with this assurance is the correspondent node able to accept Binding Updates from the mobile node which would then instruct the correspondent node to direct that mobile node's data traffic to its claimed care-of address.



Handover

A handover (HO) is the process during which a mobile node (MN) creates a new connection and disassociates from its old one.

- Handovers can be characterized as Horizontal if they are performed between connection points using the same access technology, or

- Vertical if they are performed between access points of different technologies, a case which will be more common in future heterogeneous networks.
- In addition, handovers are considered Link Layer (L2) if they are performed between connection points belonging to the same subnet, or
- Network Layer (L3) if they are performed between different subnets and require the configuration of a different IPv6 address.

Stateless Auto-configuration

The stateless address autoconfiguration mechanism allows a host to generate its own addresses in the following way.

- Access routers advertise prefixes that identify the subnet(s) associated with a link, while hosts generate an 'interface identifier' that uniquely identifies an interface on each subnet. A global address is formed by combining the two.
- The formation of an address must be followed by the Duplicate Address Detection (DAD) procedure in order to avoid address duplication on links where stateless address autoconfiguration is used.
- During DAD, the host transmits a Neighbor Solicitation for the tentative link local address and waits for some specified delay till it considers the address unique.

L2 Handover

- The IEEE 802.11 handover procedure is composed of three distinct phases: scanning, authentication, and reassociation phase.
- The scanning results in a list of APs that have been detected and it includes the related information for each detected AP, such as ESSID, the AP's MAC address, and the measured signal strength of each AP.
- After that, the MN initiates the authentication procedure by transmitting the frames related to it. If the authentication phase is successful, the MN tries to re-associate with the AP by sending a reassociation request message to the AP.

MIPv6 handover equation

- The handover delay for MIPv6 can analytically be computed as:

$$D_{MIPv6} = D_{L2} + D_{RD} + D_{DAD} + D_{REG}$$

$$D_{MIPv6} = (T_{PRB} + T_{AUTH} + T_{RASS}) + (T_{RSOL} + T_{RADV}) + D_{DAD} + (T_{HBU} + T_{HBA} + 2T_{HOTI} + 2T_{HOT} + T_{CBU} + T_{CBA})$$